AMRITA VISHWA VIDYAPEETHAM
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SCHOOL OF COMPUTING CHENNAI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #6

Activity Name: ADC Module Interfacing

Name of Student:	**Chetan Kumar G**
Registration No.:	**CH.SC.U4CSE24109**
Activity Date:	**02|10|2026**
Submission Date:	**02|10|2026**

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	**10**	

Faculty In-charge

Write an embedded C program for ARM architecture to drive ADC module. Debug and verify the program using IDE on a development computer system.

PROGRAM:

```
#include <LPC21xx.H>

/* LCD control pins */
#define RS  (1 << 0)       // P0.0
#define RW  (1 << 1)       // P0.1
#define EN  (1 << 2)       // P0.2

/* ADC registers */
#define AD0CR  (*((volatile unsigned long *)0xE0034000))
#define AD0GDR (*((volatile unsigned long *)0xE0034004))

void delay(unsigned int t)
{
    unsigned int i;
    for(i = 0; i < t; i++);
}

/* Send command to LCD */
void lcd_cmd(unsigned char cmd)
{
    IO1PIN = (IO1PIN & 0x00FFFFFF) | ((unsigned long)cmd <<
24);

    IO0CLR = RS;
    IO0CLR = RW;
    IO0SET = EN;

    delay(1000);

    IO0CLR = EN;

    delay(5000);
}

/* Send data to LCD */
void lcd_data(unsigned char data)
{
    IO1PIN = (IO1PIN & 0x00FFFFFF) | ((unsigned long)data <<
24);

    IO0SET = RS;
```

```
    IO0CLR = RW;
    IO0SET = EN;

    delay(1000);

    IO0CLR = EN;

    delay(5000);
}

/* LCD initialization */
void lcd_init(void)
{
    lcd_cmd(0x38);      // 8-bit, 2-line, 5x7
    lcd_cmd(0x0C);      // Display ON, cursor OFF
    lcd_cmd(0x01);      // Clear display
    lcd_cmd(0x06);      // Increment cursor
    delay(10000);
}

/* Display string */
void lcd_string(char *str)
{
    while(*str)
    {
        lcd_data(*str);
        str++;
    }
}

/* Display integer */
void lcd_number(unsigned int num)
{
    char buffer[6];
    int i = 0;

    if(num == 0)
    {
        lcd_data('0');
        return;
    }

    while(num > 0)
    {
        buffer[i] = (num % 10) + '0';
```

```
        num = num / 10;
        i++;
    }

    while(i > 0)
    {
        i--;
        lcd_data(buffer[i]);
    }
}

/* ADC read function */
unsigned int ADC_Read(void)
{
    unsigned int value;

    /*
       Select AD0.0
       CLKDIV = 4
       ADC enabled
    */
    AD0CR = (1 << 0) | (4 << 8) | (1 << 21);

    /* Start conversion */
    AD0CR |= (1 << 24);

    /* Wait until conversion completes */
    while((AD0GDR & (1UL << 31)) == 0);

    /* Extract 10-bit result */
    value = (AD0GDR >> 6) & 0x3FF;

    /* Stop conversion */
    AD0CR &= ~(7 << 24);

    return value;
}

int main(void)
{
    unsigned int adc_value;

    /* P0.27 = AD0.0
       Bits 22:23 of PINSEL1 = 01
    */
```

```
    PINSEL1 &= ~(3UL << 22);
    PINSEL1 |=  (1UL << 22);

    /* LCD control pins P0.0, P0.1, P0.2 as output */
    IO0DIR |= RS | RW | EN;

    /* LCD data pins P1.24 - P1.31 as output */
    IO1DIR |= 0xFF000000;

    /* Initialize LCD */
    lcd_init();

    /* Display heading */
    lcd_cmd(0x80);
    lcd_string("ADC VALUE:");

    while(1)
    {
        /* Read ADC */
        adc_value = ADC_Read();

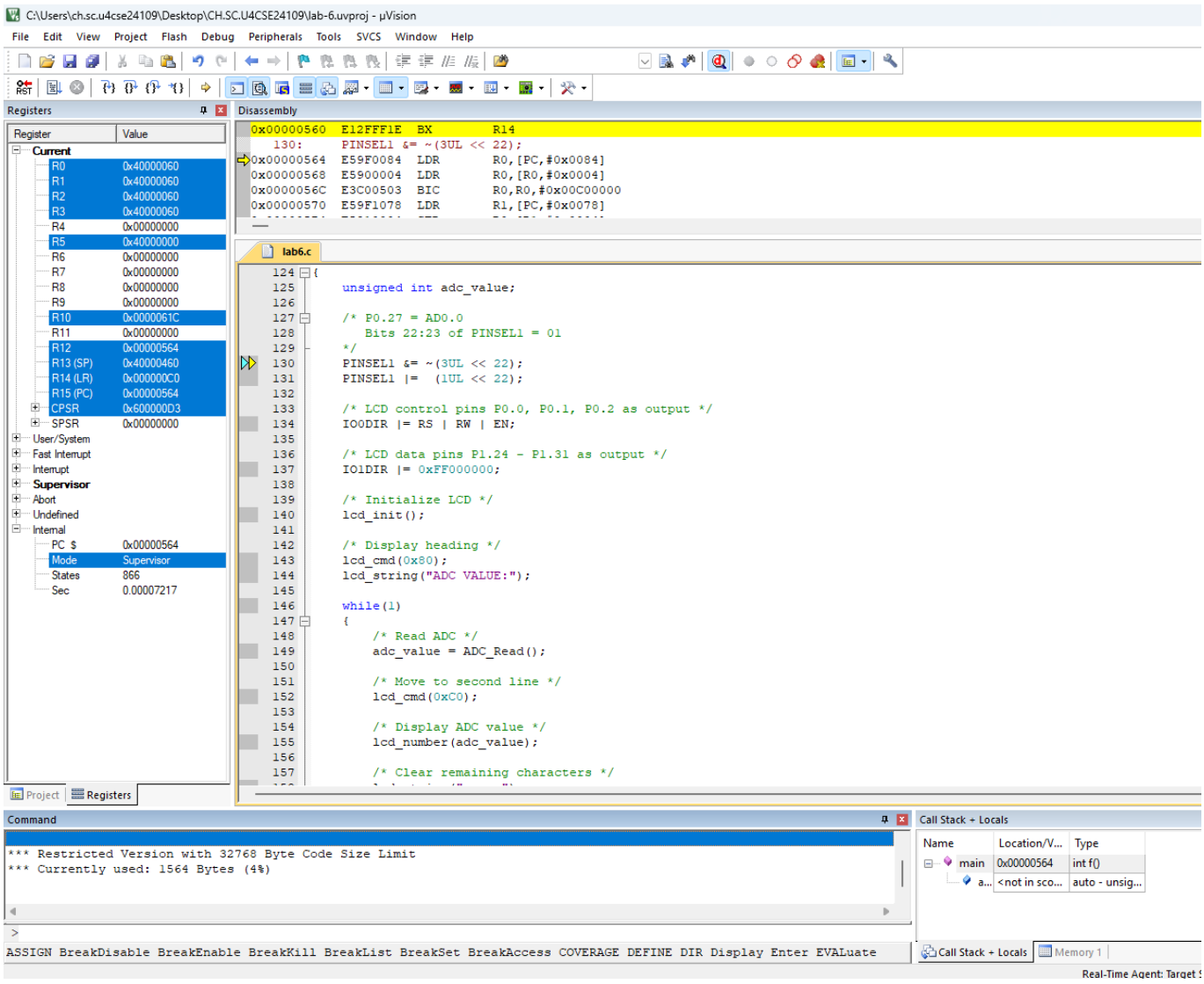
        /* Move to second line */
        lcd_cmd(0xC0);

        /* Display ADC value */
        lcd_number(adc_value);

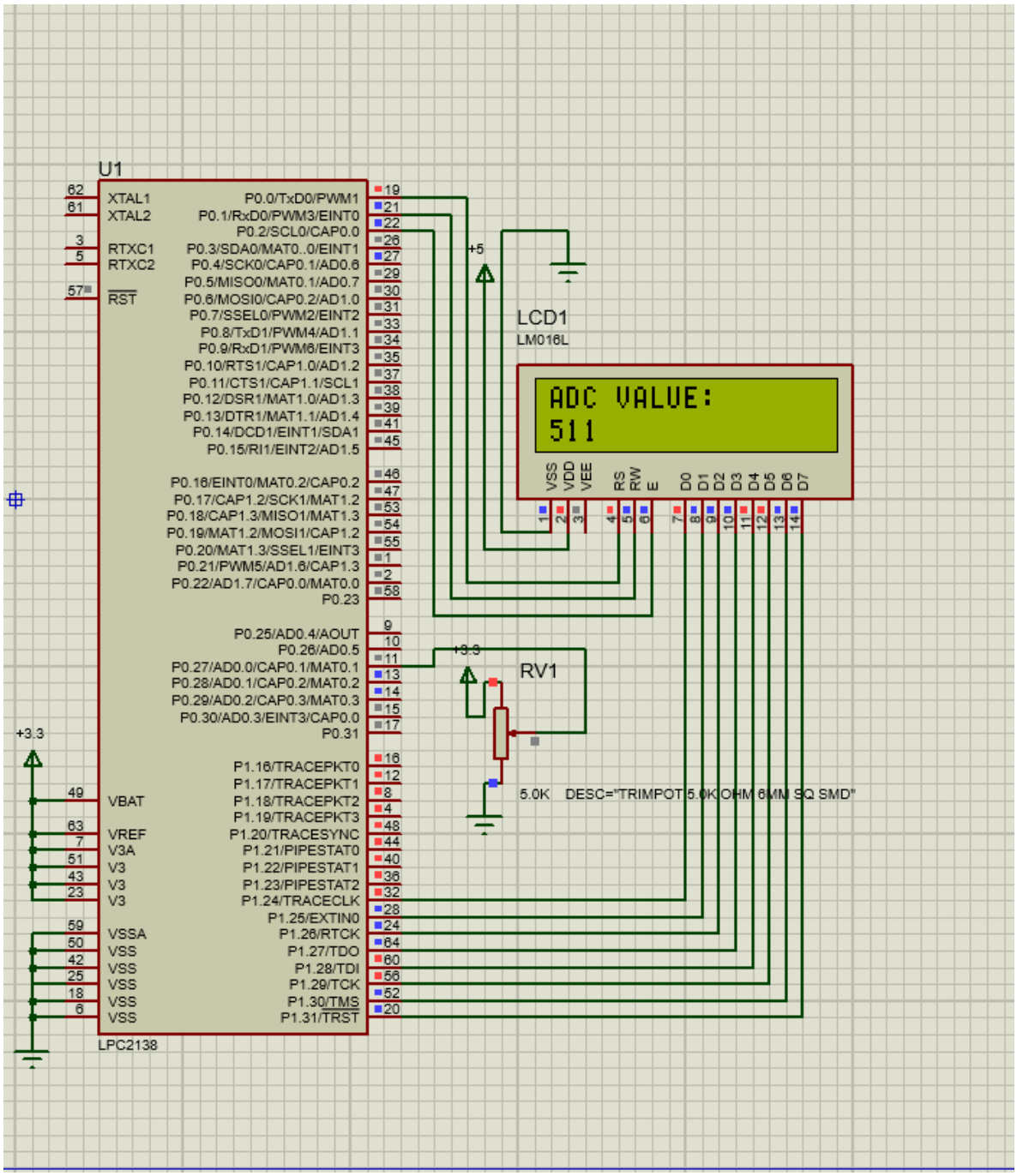
        /* Clear remaining characters */
        lcd_string("     ");

        delay(50000);
    }
}
```

OUTPUT WINDOW:


C:\Users\ch.sc.u4cse24109\Desktop\CH.SC.U4CSE24109\lab-6.uvproj - µVision
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Current
R0 0x40000060
R1 0x40000060
R2 0x40000060
R3 0x40000060
R4 0x00000000
R5 0x40000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x0000061C
R11 0x00000000
R12 0x00000564
R13 (SP) 0x40000460
R14 (LR) 0x000000C0
R15 (PC) 0x00000564
CPSR 0x600000D3
SPSR 0x00000000
User/System
Fast Interrupt
Interrupt
Supervisor
Abort
Undefined
Internal
PC $ 0x00000564
Mode Supervisor
States 866
Sec 0.00007217
Project
Registers
Disassembly
0x00000560 E12FFF1E BX R14
130: PINSEL1 &= ~(3UL << 22);
0x00000564 E59F0084 LDR R0,[PC,#0x0084]
0x00000568 E5900004 LDR R0,[R0,#0x0004]
0x0000056C E3C00503 BIC R0,R0,#0x00C00000
0x00000570 E59F1078 LDR R1,[PC,#0x0078]
lab6.c
124 {
125 unsigned int adc_value;
126
127 /* P0.27 = AD0.0
128 Bits 22:23 of PINSEL1 = 01
129 */
130 PINSEL1 &= ~(3UL << 22);
131 PINSEL1 |= (1UL << 22);
132
133 /* LCD control pins P0.0, P0.1, P0.2 as output */
134 IO0DIR |= RS | RW | EN;
135
136 /* LCD data pins P1.24 - P1.31 as output */
137 IO1DIR |= 0xFF000000;
138
139 /* Initialize LCD */
140 lcd_init();
141
142 /* Display heading */
143 lcd_cmd(0x80);
144 lcd_string("ADC VALUE:");
145
146 while(1)
147 {
148 /* Read ADC */
149 adc_value = ADC_Read();
150
151 /* Move to second line */
152 lcd_cmd(0xC0);
153
154 /* Display ADC value */
155 lcd_number(adc_value);
156
157 /* Clear remaining characters */
Command
*** Restricted Version with 32768 Byte Code Size Limit
*** Currently used: 1564 Bytes (4%)
ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE DEFINE DIR Display Enter EVALuate
Call Stack + Locals
Name Location/V... Type
main 0x00000564 int f()
a... <not in sco... auto - unsig...
Call Stack + Locals
Memory 1
Real-Time Agent: Target

OUTPUT:


U1
XTAL1
XTAL2
RTXC1
RTXC2
RST
P0.0/TxD0/PWM1
P0.1/RxD0/PWM3/EINT0
P0.2/SCL0/CAP0.0
P0.3/SDA0/MAT0..0/EINT1
P0.4/SCK0/CAP0.1/AD0.6
P0.5/MISO0/MAT0.1/AD0.7
P0.6/MOSI0/CAP0.2/AD1.0
P0.7/SSEL0/PWM2/EINT2
P0.8/TxD1/PWM4/AD1.1
P0.9/RxD1/PWM6/EINT3
P0.10/RTS1/CAP1.0/AD1.2
P0.11/CTS1/CAP1.1/SCL1
P0.12/DSR1/MAT1.0/AD1.3
P0.13/DTR1/MAT1.1/AD1.4
P0.14/DCD1/EINT1/SDA1
P0.15/RI1/EINT2/AD1.5
P0.16/EINT0/MAT0.2/CAP0.2
P0.17/CAP1.2/SCK1/MAT1.2
P0.18/CAP1.3/MISO1/MAT1.3
P0.19/MAT1.2/MOSI1/CAP1.2
P0.20/MAT1.3/SSEL1/EINT3
P0.21/PWM5/AD1.6/CAP1.3
P0.22/AD1.7/CAP0.0/MAT0.0
P0.23
P0.25/AD0.4/AOUT
P0.26/AD0.5
P0.27/AD0.0/CAP0.1/MAT0.1
P0.28/AD0.1/CAP0.2/MAT0.2
P0.29/AD0.2/CAP0.3/MAT0.3
P0.30/AD0.3/EINT3/CAP0.0
P0.31
P1.16/TRACEPKT0
P1.17/TRACEPKT1
P1.18/TRACEPKT2
P1.19/TRACEPKT3
P1.20/TRACESYNC
P1.21/PIPESTAT0
P1.22/PIPESTAT1
P1.23/PIPESTAT2
P1.24/TRACECLK
P1.25/EXTIN0
P1.26/RTCK
P1.27/TDO
P1.28/TDI
P1.29/TCK
P1.30/TMS
P1.31/TRST
VBAT
VREF
V3A
V3
V3
V3
VSSA
VSS
VSS
VSS
VSS
VSS
LPC2138
+3.3
+5
LCD1
LM016L
ADC VALUE:
511
VSS
VDD
VEE
RS
RW
E
D0
D1
D2
D3
D4
D5
D6
D7
RV1
5.0K
DESC="TRIMPOT 5.0K OHM 6MM SQ SMD"